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### PACKAGE WITH TILTED INTERFACE BETWEEN DEVICE DIE AND ENCAPSULATING MATERIAL

#### Abstract

A method includes forming a polymer layer covering a metal via in a wafer, grooving the wafer to form a trench, wherein the trench extends from a top surface of the polymer layer into the wafer, and performing a die-saw on the wafer to separate the wafer into a plurality of device dies. A kerf passes through the trench. One of the device dies is placed over a carrier. An encapsulating material is dispensed over and around the device die. The method further includes pressing and curing the encapsulating material. After the encapsulating material is cured, a sidewall of the polymer layer is tilted. A planarization is performed on the encapsulating material until the polymer layer and the metal via are exposed. A redistribution line is formed over and electrically coupled to the metal via.

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## Background/Summary

**PRIORITY CLAIM AND CROSS-REFERENCE [0001]** This application is a continuation of U.S. patent application Ser. No. 17/659,575, entitled “Package with Tilted Interface Between Device Die and Encapsulating Material,” filed on Apr. 18, 2022, which is a continuation of U.S. patent application Ser. No. 16/983,419, entitled “Package with Tilted Interface Between Device Die and Encapsulating Material,” filed on Aug. 3, 2020, now U.S. Pat. No. 11,322,419, issued May 3, 2022, which is a continuation of U.S. patent application Ser. No. 16/223,783, entitled “Package with Tilted Interface between Device Die and Encapsulating Material,” filed on Dec. 18, 2018, now U.S. Pat. No. 10,734,299, issued Aug. 4, 2020, which is a continuation of U.S. patent application Ser. No. 15/924,916, entitled “Package with Tilted Interface between Device Die and Encapsulating Material,” filed Mar. 19, 2018, now U.S. Pat. No. 10,163,745, issued Dec. 25, 2018, which is a divisional of U.S. patent application Ser. No. 15/254,472, entitled “Package with Tilted Interface between Device Die and Encapsulating Material,” filed Sep. 1, 2016, now U.S. Pat. No. 9,922,895, issued Mar. 20, 2018, which claims the benefit of the U.S. patent provisional application Ser. No. 62/332,252, filed May 5, 2016, and entitled “Package with Tilted Interface between Device Die and Encapsulating Material,” which applications are hereby incorporated herein by reference.

## BACKGROUND

**[0002]** With the evolving of semiconductor technologies, semiconductor chips/dies are becoming

increasingly smaller. In the meantime, more functions need to be integrated into the semiconductor dies. Accordingly, the semiconductor dies need to have increasingly greater numbers of I/O pads packed into smaller areas, and the density of the I/O pads rises over time. As a result, the packaging of the semiconductor dies becomes more difficult, which adversely affects the yield of the packaging.

[0003] Conventional package technologies can be divided into two categories. In the first category, dies on a wafer are packaged before they are sawed. This packaging technology has some advantageous features, such as a greater throughput and a lower cost. Further, less underfill or molding compound is needed. However, this packaging technology also suffers from drawbacks. Since the sizes of the dies are becoming increasingly smaller, and the respective packages can only be fan-in type packages, in which the I/O pads of each die are limited to a region directly over the surface of the respective die. With the limited areas of the dies, the number of the I/O pads is limited due to the limitation of the pitch of the I/O pads. If the pitch of the pads is to be decreased, solder bridges may occur. Additionally, under the fixed ball-size requirement, solder balls must have a certain size, which in turn limits the number of solder balls that can be packed on the surface of a die.

[0004] In the other category of packaging, dies are sawed from wafers before they are packaged. An advantageous feature of this packaging technology is the possibility of forming fan-out packages, which means the I/O pads on a die can be redistributed to a greater area than the die, and hence the number of I/O pads packed on the surfaces of the dies can be increased. Another advantageous feature of this packaging technology is that “known-good-dies” are packaged, and defective dies are discarded, and hence cost and effort are not wasted on the defective dies.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

[0005] Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

[0006] FIGS. **1** through **17B** illustrate the cross-sectional views of intermediate stages in the formation of fan-out packages in accordance with some embodiments.

[0007] FIG. **18** illustrates a process flow for forming a package in accordance with some embodiments.

### DETAILED DESCRIPTION

[0008] The following disclosure provides many different embodiments, or examples, for implementing different features of the invention. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

[0009] Further, spatially relative terms, such as “underlying,” “below,” “lower,” “overlying,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative

terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

[0010] A fan-out package and the method of forming the package are provided in accordance with various exemplary embodiments. Some variations of the embodiments are discussed. Throughout the various views and illustrative embodiments, like reference numbers are used to designate like elements.

[0011] FIGS. **1** through **17B** illustrate the cross-sectional views of intermediate stages in the formation of packages in accordance with some embodiments. The steps shown in FIG. **1** through **17B** are also illustrated schematically in the process flow **200** as shown in FIG. **18**.

[0012] Referring to FIG. **1**, wafer **2** is provided. Wafer **2** includes substrate **10**, which may be a semiconductor substrate such as a silicon substrate, while it may be formed of other semiconductor materials such as silicon germanium, silicon carbon, III-V compound semiconductor materials, or the like. Semiconductor devices **12**, which may be transistors, capacitors, resistors, diodes, or the like, may be formed at a surface of substrate **10**. Interconnect structure **14**, which includes metal lines and vias (not shown) formed therein, is formed over substrate **10**. The metal lines and vias may be formed of copper or copper alloys, and may be formed using damascene processes. The metal lines and vias are electrically coupled to semiconductor devices **12**. Interconnect structure **14** may include Inter-Layer Dielectric (ILD) **16** and Inter-Metal Dielectrics (IMDs) **18**, wherein contact plugs (such as source/drain plugs and gate contact plugs) are formed in ILD **16**, and the metal lines and vias are formed in IMDs **18**. In accordance with alternative embodiments, wafer **2** is an interposer wafer, and is substantially free from integrated circuit devices including transistors, resistors, capacitors, inductors, and/or the like, formed therein.

[0013] Metal pads **20** are formed over interconnect structure **14**. Metal pads **20** may include aluminum (Al), copper (Cu), silver (Ag), gold (Au), nickel (Ni), tungsten (W), alloys thereof, and/or multi-layers thereof. Metal pads **20** may be electrically coupled to semiconductor devices **12**, for example, through the metal lines, vias, and contact plugs in the underlying interconnect structure **14**. Passivation layer **22** is formed to cover edge portions of metal pads **20**. In accordance with some exemplary embodiments, passivation layer **22** includes a silicon oxide layer and a silicon nitride layer over the silicon oxide layer, although other dielectric materials may be used. An opening is formed in passivation layer **22** to expose the underlying metal pads **20**.

[0014] Polymer layer **24** is formed over passivation layer **22**, wherein polymer layer **24** extends into the openings in passivation layer **22**. Polymer layer **24** may be formed of polybenzoxazole (PBO), benzocyclobutene (BCB), polyimide, or the like. Openings are formed in polymer layer **24** to expose metal pads **20**.

[0015] Metal vias **26** are formed to extend into polymer layer **24**, and are in contact with metal pads **20**. The respective formation step is shown as step **202** in the process flow shown in FIG. **18**. Metal vias **26** may be formed of copper, aluminum, nickel, alloys thereof, and/or multi-layers thereof. In accordance with some embodiments of the present disclosure, the formation of metal vias **26** includes patterning polymer layer **24** to form openings, through which metal pads **20** are exposed. A seed layer (not shown) is then formed over, and extending into the openings of, polymer layer **24**. The seed layer may be formed of a barrier/adhesion layer comprising titanium, titanium nitride, tantalum, tantalum nitride, or the like, and a copper or copper alloy layer over the barrier/adhesion layer. A photo resist (not shown) is then formed over the seed layer and then patterned, followed by a plating process to form metal vias **26**. The photo resist is then removed. The portions of the seed layer previously covered by the photo resist are then etched, leaving metal vias **26**. Metal vias **26** are electrically coupled to integrated circuit devices **12** through metal pads **20** and the metal lines and vias in interconnect structure **14**.

[0016] Polymer layer **28** is then formed to cover and protect metal vias **26**. The respective step is

shown as step **204** in the process flow shown in FIG. **18**. In accordance with some embodiments of the present disclosure, polymer layer **28** is formed of PBO, polyimide, BCB, or the like. Polymer layer **28** may be formed of a material the same as, or different from, the material of polymer layer **24**. In accordance with some embodiments, polymer layer **28** is formed of a material that is softer than the material of polymer layer **24**.

[0017] A backside grinding is then performed to reduce the thickness of wafer **2**, and the resulting structure is shown in FIG. **2**. The respective step is shown as step **206** in the process flow shown in FIG. **18**. The backside grinding may be performed, for example, by attaching the top side of wafer **2** to a carrier (not shown), and performing a mechanical grinding or Chemical Mechanical Polish (CMP) on the backside of substrate **10**. The thickness of substrate **10** may be reduced to about 20 microns to several hundred microns, for example.

[0018] Referring to FIG. **3A**, wafer **2** is grooved. The respective step is shown as step **208** in the process flow shown in FIG. **18**. The grooving may be performed by projecting a laser beam on wafer **2** to burn-out some portions of wafer **2**. The grooving results in trench **34** in scribe line **30**, which separates two neighboring rows or columns of the chips in wafer **2**. In the top view of wafer **2**, there is a plurality of trenches **34** formed, wherein each of the scribe lines **34** of wafer **2** has a trench the same as trench **34** formed in. The trenches thus form a grid pattern in the top view. The plurality of trenches has cross-sectional views similar to the illustrated and discussed trench **34**, and is not shown separately.

[0019] Trench **34** penetrates through polymer layers **24** and **28**, and may further penetrate through interconnect structure **14** to reach substrate **10**. Accordingly, a top surface of substrate **10**, which is recessed by the laser beam, is exposed to trench **34**. Trench **34** may further extend to an intermediate level between the top surface and the bottom surface of substrate **10**.

[0020] In accordance with some embodiments, trench **34** has tilted sidewalls **36**, which are formed by tilting the projecting directions of laser beam to form the desirable tilting angle. The tilting angle  $\theta 1$  may be in the range between about 75 degrees and about 85 degrees, for example, although different angles may be adopted. Since the tilting angle  $\theta 1$  is caused by the tilting of the laser beam, the tilting angle of the laser beam is the same as tilting angle  $\theta 1$ . In accordance with some embodiments, two laser beam scans tilting to opposite directions are performed to form two tilted sidewalls **36** tilting in the opposite directions.

[0021] FIG. **3B** illustrates the grooved wafer **2** in accordance with alternative embodiments, wherein edges **36** of trench **34** are vertical or substantially vertical (for example, with tilting angle  $\theta 2$  being between about 88 degrees and 90 degrees. The trench **34** having the vertical sidewalls may be formed by projecting the laser beam in the vertical direction perpendicular to the top surface of substrate **10**.

[0022] FIG. **4** illustrates the singulation (die-saw) of wafer **2** into a plurality of chips/device dies **32**. The respective step is shown as step **210** in the process flow shown in FIG. **18**. The singulation may be performed, for example, by using blade **35** to cut through scribe lines **30**. In accordance with some embodiments, width **W1** of trench **34** is in the range between about 40  $\mu\text{m}$  and about 50  $\mu\text{m}$ . Width **W2** of the kerves caused by the singulation may be in the range between about 30  $\mu\text{m}$  and about 35  $\mu\text{m}$ . Advantageously, in the grooving, a plurality of layers such as layers **28**, **24**, **22**, **18**, and **16** is pre-grooved. Furthermore, since width **W2** of the kerves is smaller than width **W1** of trench **34**, in the singulation, the blade does not cut through the already grooved layers, and hence layers **28**, **24**, **22**, **18**, and **16** will not be peeled/delaminated by the blade. The resulting device dies **32** may include logic dies such as Central Processing Unit (CPU) dies, Graphic Processing Unit (GPU) dies, mobile application dies, or the like.

[0023] FIGS. **5** through **17A** illustrate the intermediate stages in the packaging of device die **32** in accordance with some embodiments of the present disclosure. Referring to FIG. **5**, carrier **40** is provided, and adhesive layer **42** is disposed over carrier **40**. Carrier **40** may be a blank glass carrier, a blank ceramic carrier, or the like, and may have a shape of a semiconductor wafer with a round

top-view shape. Carrier **40** is sometimes referred to as a carrier wafer. Adhesive layer **42** may be formed of a Light-to-Heat Conversion (LTHC) material, for example, although other types of adhesives may be used. In accordance with some embodiments of the present disclosure, adhesive layer **42** is capable of decomposing under the heat of light, and hence can release carrier **40** from the structure formed thereon.

[0024] Dielectric layer **44** is formed over adhesive layer **42**. In accordance with some embodiments of the present disclosure, dielectric layer **44** is a polymer layer, which may be formed of a photo-sensitive polymer such as polybenzoxazole (PBO), polyimide, or the like. In accordance with some embodiments, dielectric layer **44** is formed of a nitride such as silicon nitride, an oxide such as silicon oxide, PhosphoSilicate Glass (PSG), BoroSilicate Glass (BSG), Boron-doped PhosphoSilicate Glass (BPSG), or the like.

[0025] FIGS. **6** through **8** illustrate the formation of conductive/metal posts. The respective step is shown as step **212** in the process flow shown in FIG. **18**. Referring to FIG. **6**, conductive seed layer **50** is formed over dielectric layer **44**, for example, through Physical Vapor Deposition (PVD). Conductive seed layer **50** may be a metal seed layer including copper, aluminum, titanium, alloys thereof, or multi-layers thereof. In accordance with some embodiments of the present disclosure, conductive seed layer **50** includes a first metal layer such as a titanium layer (not shown) and a second metal layer such as a copper layer (not shown) over the first metal layer. In accordance with alternative embodiments of the present disclosure, conductive seed layer **50** includes a single metal layer such as a copper layer, which may be formed of substantially pure copper or a copper alloy.

[0026] Mask layer **52** (such as a photo resist) is applied over conductive seed layer **50**, and is then patterned using a photo lithography mask. In accordance with some embodiments of the present disclosure, photo resist **52** is a dry film, which is laminated onto conductive seed layer **50**. In accordance with alternative embodiments, photo resist **52** is formed by spin coating. As a result of the patterning (exposure and development), openings **54** are formed in photo resist **52**, through which some portions of conductive seed layer **50** are exposed.

[0027] As shown in FIG. **7**, conductive posts **56** are formed in openings **54** through plating, which may be electro plating or electro-less plating. Conductive posts **56** are plated on the exposed portions of conductive seed layer **50**. Conductive posts **56** may be metal posts formed of copper, aluminum, tungsten, nickel, or alloys thereof.

[0028] After the plating of conductive posts **56**, photo resist **52** is removed. As a result, the portions of conductive seed layer **50** that are previously covered by photo resist **52** are exposed. Next, an etching step is performed to remove the exposed portions of conductive seed layer **50**, wherein the etching may be an anisotropic or isotropic etching. The portions of conductive seed layer **50** that are overlapped by conductive posts **56**, on the other hand, remain not etched. Throughout the description, the remaining underlying portions of conductive seed layer **50** are referred to as the bottom portions of conductive posts **56**. The resulting structure is shown in FIG. **8**. In FIG. **8** and subsequent drawings, the remaining portions of conductive seed layer **50** are considered as parts of conductive posts **56**, and are not shown separately.

[0029] FIG. **9** illustrates the placement of device die **32** over dielectric layer **44** and carrier **40**. The respective step is shown as step **214** in the process flow shown in FIG. **18**. Device die **32** may be attached to dielectric layer **44** through a die attach film (not shown). The die attach film may be adhered to the bottom surface of wafer **2** (FIG. **4**) before the singulation, and then sawed along with wafer **2** in the singulation step. As a result, the edges of the die attach film are co-terminus as device die **32**. It is appreciated that the packaging is performed at the wafer level, and although there is one device die **32** illustrated, a plurality of placed device dies identical to device die **32** is actually placed over dielectric layer **44**, wherein the plurality of placed device dies is arranged as an array including a plurality of rows and a plurality of columns.

[0030] Referring to FIG. **10**, encapsulating material **60** is dispensed on device die **32** and conductive posts **56**. The respective step is shown as step **216** in the process flow shown in FIG.

**18.** Encapsulating material **60** fills the gaps between device die **32** and conductive posts **56**, and may be in contact with dielectric layer **44**. Encapsulating material **60** may include a molding compound, a molding underfill, an epoxy, or a resin. Encapsulating material **60** may include a polymer-based material and filler particles, which may be formed of silicon oxide, aluminum oxide, or the like. The top surface of encapsulating material **60** is higher than the top ends of metal vias **26** and conductive posts **56**.

[0031] Referring to FIG. **11**, encapsulating material **60** is compressed and cured. The respective step is shown as step **218** in the process flow shown in FIG. **18**. The compression may be performed by using top mold **62** and release film **64** to push encapsulating material **60**. The pushing force is represented by arrow **66**. Through the compression, encapsulating material **60** is spread more uniformly, so that no void is formed in encapsulating material **60**. During the compression, a mold (not shown) surrounds, and may be underlying, carrier **40** to hold encapsulating material **60**. During the compression, encapsulating material **60** is cured, for example, by a heating.

[0032] As shown in FIG. **11**, slanted sidewalls **36'** are generated. There are several reasons that may cause slanted sidewalls **36'** to have the profile as shown in FIG. **11**. For example, the tilted sidewalls **36'** receive the downward pressing force, and hence the downward force is partially converted to lateral force due to the tilting of sidewalls **36** (FIG. **4**). Also, the compression of encapsulating material **60** also contributes to the lateral force. The top layers **24** and **28** are polymer layers, which are soft, and hence yield to the lateral force pushing these layers toward the center line of device die **32**. Furthermore, the sidewalls of layers **24** and **28**, which are portions of sidewalls **36'**, are curved. In accordance with some embodiments, each of the sidewalls of layers **24** and **28** is continuously curved with no abrupt changes in the slope therein. There may be, or may not be, an abrupt change in slopes at the interfaces between layers **24** and **28**. When being compressed, encapsulating material **60** is cured, and the profile as shown in FIG. **11** is fixed. Mold **62** and release film **64** are then removed, as shown in FIG. **12**.

[0033] Next, a planarization step such as a CMP step or a grinding step is performed to planarize encapsulating material **60**, until conductive posts **56** and metal vias **26** are exposed. The respective step is shown as step **220** in the process flow shown in FIG. **18**. The resulting structure is shown in FIG. **13**. Metal vias **26** of device die **32** are also exposed as a result of the planarization. Due to the planarization, the top surfaces of conductive posts **56** are substantially level (coplanar) with the top surfaces of metal vias **26**, and are substantially level (coplanar) with the top surface of encapsulating material **60**. Due to the planarization, some spherical filler particles (not shown) in encapsulating material **60** have their top portions removed, and hence leaving the filler particles with planar top surfaces and rounded sidewalls and bottom surfaces.

[0034] Referring to FIG. **14**, one or more layers of dielectric layers **68** and the respective Redistribution Lines (RDLs) **70** are formed over encapsulating material **60**, conductive posts **56**, and metal vias **26**. The respective step is shown as step **222** in the process flow shown in FIG. **18**. RDLs **70** are referred to as front side RDLs since they are on the front side of device die **32**. In accordance with some embodiments of the present disclosure, dielectric layers **68** are formed of a polymer(s) such as PBO, polyimide, or the like. In accordance with alternative embodiments of the present disclosure, dielectric layers **68** are formed of an inorganic dielectric material(s) such as silicon nitride, silicon oxide, silicon oxynitride, or the like.

[0035] RDLs **70** are formed to electrically couple to metal vias **26** and conductive posts **56**. RDLs **70** may also interconnect metal vias **26** and conductive posts **56** with each other. RDLs **70** may include metal traces (metal lines) and vias underlying and connected to the metal traces. In accordance with some embodiments of the present disclosure, RDLs **70** are formed through plating processes, wherein each layer of RDLs **70** includes a seed layer (not shown) and a plated metallic material over the seed layer. The seed layer and the plated metallic material may be formed of the same material or different materials. Under-Bump Metallurgies (UBMs) **72** are then formed to

extend into the top dielectric layer **68** and in contact with the metal pads in the top RDLs **70**.

[0036] As shown in FIG. **15**, electrical connectors **76** are formed on UBMs **72**. The formation of electrical connectors **76** may include placing solder balls over RDLs **70** and then reflowing the solder balls. In accordance with alternative embodiments of the present disclosure, the formation of electrical connectors **76** includes performing a plating step to form solder regions over RDLs **70** and then reflowing the solder regions. Electrical connectors **76** may also include metal pillars, or metal pillars and solder caps, which may also be formed through plating. Throughout the description, the combined structure including device die **32**, conductive posts **56**, encapsulating material **60**, RDLs **70**, and dielectric layers **68** will be referred to as composite wafer **74**, which is a composite wafer including a plurality of device dies **32**. Carrier **40** (FIG. **14**) may then be de-bonded from composite wafer **74**. As also shown in FIG. **15**, surface-mount device **78**, which may be a discrete passive device such as a capacitor, a coil, a transformer, or the like, is bonded to composite wafer **74** through solder regions **80**.

[0037] Referring to FIG. **16**, openings **82** are formed in dielectric layer **44** to expose metal posts **56**. Openings **82** may be formed through laser drill, for example. A die-saw is then performed to singulate composite wafer **74** into a plurality of packages **86**, each including (at least) one of device dies **32** and the corresponding conductive posts **56**. The respective step is also shown as step **224** in the process flow shown in FIG. **18**. In accordance with some embodiments of the present disclosure, the die-saw is performed using a blade, which is rotated to cut composite wafer **74** during the die-saw. The respective step is shown as step **224** in the process flow shown in FIG. **18**.

[0038] FIG. **17A** illustrates the bonding of package components **88** and **92** to package **86**, thus forming package **94**. In accordance with some embodiments of the present disclosure, package component **88** may include device die(s), which may be memory dies such as Static Random Access Memory (SRAM) dies, Dynamic Random Access Memory (DRAM) dies, or the like. Package component **92** may be a package, a package substrate, a Printed Circuit Board (PCB), an interposer, or the like. After the bonding, underfills (not shown) may be disposed into the gaps between package **88** and package components **88** and **92**, and is then cured.

[0039] In the package **94** as shown in FIG. **17A**, the sidewalls **36'** of device die **32** includes tilted portions. For example, tilted portions **36'A** are the sidewalls of polymer layer **28**, tilted portions **36'B** are the sidewalls of polymer layer **24**, and portions **36C'** (which may be tilted, not tilted, or partially tilted) are the sidewalls of passivation layer **22**, dielectric layers **16** and **18**, and substrate **10**. Accordingly, the interface between encapsulating material **60** and device die **32** also include tilted portions.

[0040] In accordance with some embodiments, sidewall portion **36'A** of polymer layer **28** has tilt angle  $\theta_3$ , which may be in the range between about 50 degrees and about 70 degrees. The top surface of polymer layer **28** is recessed laterally (toward the vertical center line of device die **32**) the respective bottom surface by distance **X2** (also refer to FIG. **11**) in the range between about 2  $\mu\text{m}$  and about 5  $\mu\text{m}$ . Thickness **Y2** of polymer layer **28** may be in the range between about 5  $\mu\text{m}$  and about 15  $\mu\text{m}$ . Sidewall portions **36'B** of polymer layer **24** has tilt angle  $\theta_4$ , which is greater than tilt angle  $\theta_3$ . Tilt angle  $\theta_4$  may be in the range between about 70 degrees and about 85 degrees. The top surface of polymer layer **24** is recessed laterally relative to the respective bottom surface by a distance **X1** in the range between about 1  $\mu\text{m}$  and about 2  $\mu\text{m}$ . Thickness **Y1** of polymer layer **24** may be in the range between about 4  $\mu\text{m}$  and about 6  $\mu\text{m}$ .

[0041] Sidewalls **36'A** and **36'B** may also be curved. Furthermore, although there may be (or may not be) a discontinuity in the slope ratio of sidewalls **36'A** and **36'B**, the slopes of each of **36'A** and **36'B** may be continuously changed, with the upper portions of each of sidewalls **36'A** and **36'B** being increasingly more tilted than the respective lower portions.

[0042] Furthermore, distance **X1** may be greater than, smaller than, or equal to distance **X2**, depending on the thicknesses **Y1** and **Y2** of polymer layers **24** and **28**, respectively. For example, ratio **X2/X1** may be in the range between about 0.1 and about 0.5, in the range between about 0.6



and about 1, in the range between about 1 and about 2, or in the range between about 2 and about 8.

[0043] The top surface of polymer layer **28** is laterally recessed from the respective outmost edge of substrate **10** by distance **X3**, which may be in the range between about 1.0  $\mu\text{m}$  and about 1  $\mu\text{m}$ , in the range between about 1.1  $\mu\text{m}$  and about 3  $\mu\text{m}$ , in the range between about 3.1  $\mu\text{m}$  and about 5  $\mu\text{m}$ , in the range between about 5.1  $\mu\text{m}$  and about 10  $\mu\text{m}$ , or in the range between about 10.1  $\mu\text{m}$  and about 20  $\mu\text{m}$ . The top surface of polymer layer **28** is also laterally recessed from the respective edge of substrate **10** formed by grooving by distance **X4**, which may be in the range between about 1.0  $\mu\text{m}$  and about 1  $\mu\text{m}$ , in the range between about 1.1  $\mu\text{m}$  and about 3  $\mu\text{m}$ , in the range between about 3.1  $\mu\text{m}$  and about 5  $\mu\text{m}$ , in the range between about 5.1  $\mu\text{m}$  and about 10  $\mu\text{m}$ , or in the range between about 10.1  $\mu\text{m}$  and about 20  $\mu\text{m}$ . In accordance with some embodiments, value (**X3-X4**) is greater than about 0.1  $\mu\text{m}$ , and may be in the range between about 0.1  $\mu\text{m}$  and about 0.9  $\mu\text{m}$ , in the range between about 1  $\mu\text{m}$  and about 3  $\mu\text{m}$ , or in the range between about 3.1  $\mu\text{m}$  and about 20  $\mu\text{m}$ .

[0044] The surfaces of substrate **10** may form a step, which step contacts encapsulating material **60**. The step is formed of sidewall portion **36'C**, sidewall portion **36'D**, and horizontal surface **10'** of substrate **10**. The sidewall portion **36'D** (the sidewalls of substrate **10**) is vertical and perpendicular to the bottom surface of substrate **10**. Sidewall portion **36'C** may be vertical or tilted. If tilted, the tilt angle  $\theta 5$  of sidewall portions **36'C** is greater than both tilt angles  $\theta 3$  and  $\theta 4$ .

[0045] FIG. **17B** illustrates package **94** formed in accordance with some embodiments of the present disclosure, the package is similar to the package **94** shown in FIG. **17A**, except that instead of forming two polymer layers **24** and **28**, a single polymer layer **24** is formed, which extends from passivation layer **22** to dielectric layers **68**. Metal vias **26** are formed in polymer layer **24**. In accordance with some embodiments, each of sidewall **36'** includes tilted (which may be curved) portions. The details are similar to what are shown and discussed for FIG. **17A**, and may be found referring to the discussion of the corresponding features with the corresponding references numerals.

[0046] The embodiments of the present disclosure have some advantageous features. The sidewalls of device die **32** are tilted. The tilt sidewalls advantageously smoothen the downward movement of encapsulating material **60** during the compress molding, and hence reduce the stress suffered by the molded device die. Furthermore, the tilted sidewalls make it easy for encapsulating material **60** to be compressed aside of device die **32** instead of being pressed directly onto device die **32**, thus reduce the deformation of device die **32** in the compression of the encapsulating material.

[0047] In accordance with some embodiments of the present disclosure, a method includes forming a polymer layer covering a metal via in a wafer, grooving the wafer to form a trench, wherein the trench extends from a top surface of the polymer layer into the wafer, and performing a die-saw on the wafer to separate the wafer into a plurality of device dies. A kerf passes through the trench. One of the device dies is placed over a carrier. An encapsulating material is dispensed over and around the device die. The method further includes pressing and curing the encapsulating material. After the encapsulating material is cured, a sidewall of the polymer layer is tilted. A planarization is performed on the encapsulating material until the polymer layer and the metal via are exposed. A redistribution line is formed over and electrically coupled to the metal via.

[0048] In accordance with some embodiments of the present disclosure, a method includes performing a grooving on a wafer to form a plurality of trenches extending from a top surface of the wafer to an intermediate level of the wafer, and performing a die-saw on the wafer to separate the wafer into a plurality of device dies. Kerves of the die-saw pass through respective ones of the plurality of trenches, and the kerves are narrower than respective ones of the plurality of trenches. A device die in the plurality of device dies is placed over a carrier. The device die is encapsulated in an encapsulating material. After the device die is encapsulated, a sidewall of the device die is tilted. The method further includes performing a planarization on the encapsulating material until a metal via in the device die is exposed, and forming a redistribution line over and electrically coupling to the metal via.

[0049] In accordance with some embodiments of the present disclosure, a package includes a device die, which includes a substrate, and a sidewall with a tilted portion neither parallel to nor perpendicular to a bottom surface of the substrate. The package further includes an encapsulating material encapsulating the device die therein, wherein the tilted portion of the sidewall is in contact with the encapsulating material, a metal post penetrating through the encapsulating material, and redistribution lines over and electrically coupling to the metal post and device die.

[0050] In accordance with some embodiments of the present disclosure, a package includes a device die, which includes a semiconductor substrate, an interconnect structure over the semiconductor substrate, a metal pillar over and electrically coupled to the interconnect structure, and a polymer layer encircling the metal pillar. The polymer layer has a tilted sidewall, which is neither parallel to nor perpendicular to a major bottom surface of the semiconductor substrate. An encapsulating material encapsulates the device die therein. Redistribution lines are formed over and electrically coupling to the metal pillar.

[0051] The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

## Claims

1. A structure comprising: a device die comprising: a semiconductor substrate comprising: a first edge; a second edge over the first edge, wherein the second edge is laterally recessed toward a center vertical line of the device die than the first edge; and a top surface connecting the first edge to the second edge; a plurality of dielectric layers over the semiconductor substrate; a first polymer layer over the plurality of dielectric layers, wherein a third edge of the first polymer layer is further laterally recessed toward the center vertical line of the device die than the second edge; and a conductive feature comprising a part in the first polymer layer.
2. The structure of claim 1 further comprising a second polymer layer over the first polymer layer, wherein a fourth edge of the second polymer layer is further laterally recessed toward the center vertical line of the device die than the third edge of the first polymer layer.
3. The structure of claim 1, wherein the first edge and the second edge are vertical, and the third edge is slanted.
4. The structure of claim 1, wherein the first edge and the second edge are straight, and the third edge is curved.
5. The structure of claim 1, wherein lower portions of the third edge of the first polymer layer are more vertical than respective upper portions of the third edge of the first polymer layer.
6. The structure of claim 1 further comprising a molding compound in contact with the first edge, the second edge, and the third edge, wherein the molding compound further forms a horizontal interface with the top surface of the semiconductor substrate.
7. The structure of claim 1, wherein upper layers of the plurality of dielectric layers are increasingly narrower than respective lower layers of the plurality of dielectric layers.
8. The structure of claim 1 further comprising: a dielectric layer over and contacting the device die; and a redistribution line in the dielectric layer, wherein the redistribution line is electrically coupled to the conductive feature.
9. The structure of claim 1, wherein edges of the plurality of dielectric layers are slanted, with a lower edge of a lower layer of the plurality of dielectric layers being more vertical than an upper

edge of a corresponding upper layer of the plurality of dielectric layers.

**10.** The structure of claim 1, wherein the first edge, the second edge, and the top surface of the semiconductor substrate form a step.

**11.** A structure comprising: a device die comprising: a semiconductor substrate; an interconnect structure over the semiconductor substrate, wherein the interconnect structure comprises a plurality of dielectric layers; a metal pad over and electrically coupling to the interconnect structure; a passivation layer covering a portion of the metal pad; a first polymer layer over the passivation layer; a second polymer layer over the first polymer layer, wherein in a cross-sectional view of the structure, from a top surface level of the second polymer layer to a bottom surface level of one of the plurality of dielectric layers, the device die is increasingly wider; and a conductive feature over and contacting the metal pad, wherein the conductive feature comprises parts in both of the first polymer layer and the second polymer layer.

**12.** The structure of claim 11, wherein the second polymer layer has a first curved sidewall, and lower portions of the first curved sidewall are more vertical than respective upper portions of the first curved sidewall.

**13.** The structure of claim 12, wherein the first polymer layer has a second curved sidewall, and lower portions of the second curved sidewall are more vertical than respective upper portions of the second curved sidewall.

**14.** The structure of claim 12 further comprising a molding compound contacting the first curved sidewall.

**15.** The structure of claim 11, wherein the first polymer layer is in physical contact with the second polymer layer.

**16.** The structure of claim 11, wherein in the cross-sectional view, the semiconductor substrate comprises a vertical edge.

**17.** The structure of claim 16, wherein in the cross-sectional view, an edge of the device die is continuously curved, and is joined to a top surface of the device die.

**18.** A structure comprising: a device die comprising: a semiconductor substrate; dielectric layers over the semiconductor substrate, wherein the dielectric layers comprise: a plurality of inter-metal dielectrics; a polymer layer over the plurality of inter-metal dielectrics, wherein the polymer layer comprises a curved sidewall that is continuously curved; and a top surface joined to a topmost end of the curved sidewall; a molding compound, with the device die being in the molding compound; and a first dielectric layer over and contacting both of the device die and the molding compound, wherein the top surface of the device die contacts a bottom surface of the dielectric layer.

**19.** The structure of claim 18 further comprising a second dielectric layer overlapped by both of the device die and the molding compound.

**20.** The structure of claim 18, wherein the semiconductor substrate of the device die comprises a vertical edge, and the curved sidewall is laterally recessed from the vertical edge of the semiconductor substrate.

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